S^2-Differential Feature Awareness Network for Hyperspectral Image Fusion

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***Abstract*—The fusion of hyperspectral image (HSI) and multispectral image (MSI) aims to fully exploit the complementary advantages of both modalities to generate a high-resolution HSI. However, existing HSI fusion methods fail to fully utilize frequency-domain information when capturing deep correlations between spectral bands, leading to difficulties in accurately preserving spatial texture details. Furthermore, these methods often overlook the intrinsic differential characteristics in HSI fusion, which easily causes spatial–spectral (S^2) decoupling distortions. To address the aforementioned challenges, this article proposes a S^2 differential feature awareness network (SSDAN) that employs a multiscale input strategy to fully preserve the rich spatial structures in the source images. Specifically, a spatial–frequency fusion network (SFFN) with a dual-branch architecture is designed to separately model multilevel spatial–frequency features of HSI and MSI, thereby effectively enhancing the joint representation of high-frequency details and global–local structures. In addition, we introduce a differential S^2 complementarity module (DSSCM) that explicitly models modality-specific detailed features and adaptively aligns the dual-branch fused features with the shallow features, effectively mitigating decoupling distortions caused by feature mismatches. The code will be available at https://github.com/MSFLabX/SSDAN**

***Index Terms*—Differential feature, frequency, hyperspectral image (HSI), texture structure.**

I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D data cube containing rich spectral information, which provides strong support for precise material identification. Consequently, it has been widely used in many fields such as classification [1], [2], [3], [4], geological exploration [5], and precision agriculture [6]. However, due to inherent limitations in physical imaging systems, it is difficult to simultaneously obtain HSI with high spectral and high spatial resolution, which severely restricts its practical use. Current methods for acquiring high-resolution HSI can be mainly divided into two categories. The first category involves enhancing hardware to increase the photon throughput per spatial unit, but such solutions are costly and involve long development cycles. The second category focuses on generating high-resolution HSIs through algorithmic reconstruction. As a representative example, HSI fusion leverages the rich spectral information of HSIs and the high spatial details of multispectral images (MSIs) to achieve complementary information integration, thereby producing high-quality HSIs.

Extensive research on HSI–MSI fusion conducted by many researchers can be broadly classified into two main manners: traditional model-driven methods [7], [8], [9] and deep learning-based approaches [10], [11], [12], [13], [14]. Traditional methods offer strong interpretability because they explicitly model the spatial–spectral (S^2) degradation process and incorporate handcrafted priors. However, they typically require solving complex optimization problems in an iterative manner, resulting in high computational cost. Moreover, their performance is highly sensitive to manually tuned parameters and depends heavily on the accuracy of assumed priors, which often leads to limited generalization in diverse or unconstrained real-world scenes. In contrast, deep learning-based methods have rapidly advanced in recent years due to their powerful end-to-end learning capabilities [15], [16], [17], [18], [19], [20], [21]. These approaches can automatically extract S^2 features and learn nonlinear cross-modal mapping relationships directly from data, thereby achieving superior reconstruction accuracy and stronger generalization performance [22], [23]. For example, Sun et al. [22] propose a three-branch multiscale deep cross-fusion transformer network, which effectively addresses the problem of limited fusion quality caused by insufficient S^2 feature interaction. Fu et al. [23] propose a dual-branch multiscale S^2 dilated transformer network, significantly reducing computational complexity while effectively overcoming the local receptive field limitation of CNN. Although they already have very good performance, they generally overlook frequency-domain information and lack a global–local coupling mechanism, failing to well preserve the spatial texture details of images and easily leading to significant degradation of high-frequency details and edge structures in the fused results.

To address the above challenges, we propose a S^2 differential feature awareness network (SSDAN) to preserve fine-grained texture and structural details for high-fidelity reconstruction. Specifically, a multiscale input strategy is

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adopted in place of the conventional U-Net architecture to better preserve original spatial information. Building upon this design, a spatial–frequency fusion network (SFFN) is developed, which consists of the global–local frequency-domain fusion module (GLFFM), the spectral–spatial rectification block (SSRB), the differential S^2 complementarity module (DSSCM), and the high-frequency information fusion module (HFIFM) to collaboratively process multilevel features. Specifically, the GLFFM exploits global and local frequency characteristics of HSI and MSI to capture global structural information while better model local texture details. The SSRB adopts a self-attention mechanism and a spatial rectification block to enable deep interaction and rectification between spatial and spectral features. The DSSCM further refines the fused representations by leveraging the original HSI and MSI to compensate for S^2 discrepancies. Finally, to alleviate the distortion of high-frequency information in deep feature representations, the HFIFM incorporates MSI-derived high-frequency features to improve texture sharpness and detail richness in the reconstructed images.

The main contributions are summarized as follows.

1) We propose an SSDAN that employs a multiscale input strategy to fully preserve the spatial information of the original image. Building upon this architecture, the SFFN is designed to collaboratively fuse spatial and frequency-domain features while enhancing high-frequency information.
2) We design a DSSCM that adaptively enhances spatial and spectral features to mitigate S^2 decoupling distortion. This is achieved by jointly leveraging fusion results from the spatial and frequency domains, along with their respective spatial and spectral discrepancies with HSI and MSI.
3) The effectiveness of the proposed method is validated through experiments on three public datasets, where it consistently surpasses several SoTA approaches in HSI fusion performance.

The remainder of this article is organized as follows. Section II reviews related work in HSI fusion. Section III details the proposed methodology. Section IV presents qualitative and quantitative comparisons as well as ablation studies and hyperparameter experiments. Finally, Section V concludes the article.

II. Related Work

A. HSI Fusion

Traditional HSI fusion methods mainly rely on manually designed transformation rules to exploit the correlations between HSIs and MSIs. These approaches can be broadly categorized into two groups: panchromatic-sharpening-based methods and model-based methods. The pansharpening fusion strategy exploits the complementary strengths of HR-MSIs and LR-HSIs to generate hyperspectral data with both high spatial resolution and rich spectral information [24], [25]. Model-based methods typically perform fusion by formulating and iteratively solving mathematical optimization problems, in which the optimization process is constrained by manually specified priors [8], [9]. However, these approaches are highly sensitive to manually tuned parameters and heavily rely on the accuracy of the priors, which significantly limits their adaptability and generalization capability.

Driven by deep learning techniques, HSI fusion has achieved significant progress in recent years. Compared with traditional fusion approaches, these methods demonstrate superior performance owing to their strong feature extraction capability. Early studies in this field primarily investigate CNN as the dominant architectural paradigm. For example, Han et al. [26] propose two novel CNN architectures for HSI super-resolution, which address the problem that single-dimensional upsampling-based CNNs ignore the complementary information inherent in LR-HSIs and HR-MSIs. Ran et al. [27] propose a dual-branch convolutional neural network framework, called GuidedNet, which efficiently fuses LR-HSIs and HR-MSIs through multiscale guidance and residual learning, while maintaining a compact parameter size. Nonetheless, CNNs are inherently limited in capturing global context due to their restricted receptive fields. In contrast, Transformer-based methods leverage attention mechanisms to model global dependencies more effectively, achieving superior performance in HSI fusion. Ma et al. [28] achieve bidirectional information interaction between modalities through DCTransformer, which is further integrated with Swin Transformer blocks to enhance contextual self-attention. Despite their promising performance, these methods only explore correlations within the spatial domain and are thus insufficient to fully preserve the intrinsic structural information and spatial texture details of the original images.

B. Fourier Transform

The Fourier transform is extensively employed in frequency-domain analysis, as it converts time-domain signals into the frequency domain, enabling signal filtering, spectrum analysis, and other operations that are difficult to perform directly in the time domain. In recent years, the Fourier transform has been widely adopted in various computer vision tasks [29], [30], [31] owing to its ability to decompose an image into frequency components. Given the considerable potential of the Fourier transform in image processing, this technique has been extensively adopted for remote sensing image analysis. Qin et al. [32] capture spatial–frequency complementary information by mapping multimodal features into multiple fractional-order domains and dynamically fusing representations across different domains via optimal transmission mechanisms, thereby effectively exploiting frequency-domain complementarity. At present, only a limited number of methods have introduced the Fourier transform for HSI fusion. For example, Liang et al. [33] introduce a Fourier-enhanced implicit neural representation (INR) fusion network to address the problem where INR is prone to high-frequency information loss due to their inadequate global perception capability. Zhang et al. [34] develop an effective new paradigm for HSI fusion to explore the complementary fusion from both the spatial and frequency domains. However, these methods only perform a Fourier transform on the entire image, thus neglecting local frequency-domain relationships. Incorporating fusion in the

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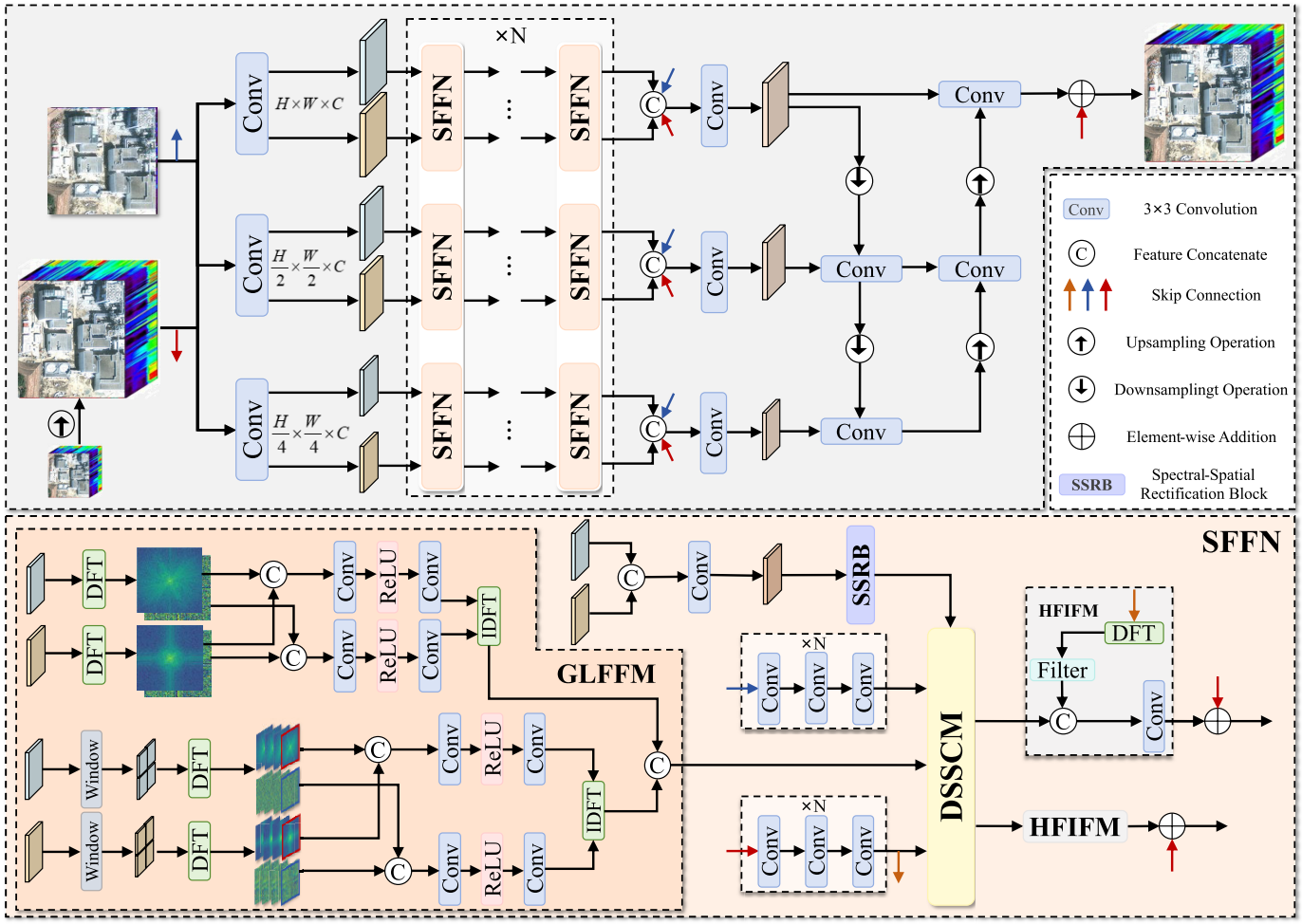


Fig. 1. Overall framework of the proposed SSDAN. The SFFN comprises four core components: the GLFFM, the SSRB, the DSSCM, and the HFIFM.

local frequency domain can better model local interactions between HSI and MSI, and help preserve spatial details. To leverage this property, we perform fusion in both the global and local frequency domains simultaneously, which can preserve the global structural information of the original input and enhance the texture detail of local regions in the fused image.

III. Methodology

In this section, we elaborate on the proposed SSDAN. First, the overall framework of SSDAN is presented, followed by a detailed description of the SFFN and its constituent components, and finally, the loss function is introduced.

A. Overall Framework

The framework of the proposed SSDAN is illustrated in Fig. 1. The overall architecture adopts a multi-input multi-output structure, which has been widely applied to HSI fusion tasks [35]. Specifically, the proposed framework is composed of the SFFN and a multiscale fusion strategy. The SFFN performs preliminary fusion of HSI and MSI in both spatial and frequency domains, while enhancing S^2 representations and high-frequency details. Subsequently, the multiscale fusion strategy further integrates the fused features across different scales to achieve high-fidelity reconstruction. Let $x \in \mathbb{R}^{H\times W\times c}$, $y \in \mathbb{R}^{h\times w\times C}$ denote HR-MSI and LR-HSI, respectively, where H and W (or h and w) represent the spatial dimensions and c (or C) is the number of spectral bands. We first upsample y to the same spatial resolution as x to obtain y_1. x_2, y_2, x_3, and y_3 are obtained by successively downsampling x and y_1 twice by bilinear interpolation with a scaling factor of $1/2$. After obtaining the three groups of features $\{x, y_1\}$, $\{x_2, y_2\}$, and $\{x_3, y_3\}$, we apply convolutional layers to expand their channel dimensions, resulting in $\{x'_1, y'_1\}$, $\{x'_2, y'_2\}$, and $\{x'_3, y'_3\}$. The process of generating three groups of features at different scales can be described as follows:

$$\begin{cases} x'_n = Conv\left(downsample^{n-1}(x)\right) \\ y'_n = Conv\left(downsample^{n-1}(upsample(y))\right) \end{cases} \tag{1}$$

where $Conv(\bullet)$ denotes a 3×3 convolutional layer, and $downsample^n$ denotes the number of downsampling operations, $n = 1, 2, 3$, and the obtained three groups of features are input to multiple SFFN, respectively. It can be expressed by the formula as follows:

$$x^e_n, y^e_n = SFFN_i\left(x'_n, y'_n\right) \tag{2}$$

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where $SFFN_i(\bullet)$ denotes the SFFN at the ith stage, and x_n^e and y_n^e are the outputs corresponding to the nth scale features. The output features are integrated using a multiscale fusion strategy. Specifically, features are upsampled or downsampled to a unified scale and fused through multiple convolutional layers, and the final reconstruction is obtained by adding the fused features to y_1.

B. Spatial–Frequency Fusion Network

We design the SFFN to better model S^2 representations while preserving fine-grained spatial details. The SFFN is constructed with four components: GLFFM, SSRB, DSSCM, and HFIFM, which are described in detail below.

1) Global–Local Frequency-Domain Fusion Module: We propose a GLFFM that effectively integrates global frequency distribution characteristics with local detail-related frequency components. In the global frequency-domain fusion part, we use the discrete Fourier transform (DFT) to obtain the amplitude and phase components of the MSI and HSI. For the input HSI features F_{hsi} and MSI features F_{msi}, the above process can be described as follows:

$$F_{hsi_amp}, F_{hsi_pha} = DFT\left(F_{hsi}\right) \tag{3}$$

$$F_{msi_amp}, F_{msi_pha} = DFT\left(F_{msi}\right). \tag{4}$$

After decomposing the spatial domain features into amplitude and phase components, we perform feature fusion by explicitly leveraging the distinct functional roles of these two components. Specifically, the amplitude components of HSI and MSI features are concatenated along the channel dimension, while the phase components are processed in the same manner. The concatenated amplitude and phase features are then fused through two independent convolutional blocks to obtain the fused amplitude F_{g_amp} and the fused phase F_{g_pha}, respectively. Each convolutional block consists of multiple 1×1 convolutional layers with LeakyReLU activation functions. F_{g_amp} and F_{g_pha} are then reconstructed in the spatial domain through the inverse DFT (IDFT) to obtain the global frequency-domain fusion result F_g. The whole process can be formulated as follows:

$$F_{g_amp} = Conv_{1\times1}\left(Cat\left(F_{msi_amp}, F_{hsi_amp}\right)\right) \tag{5}$$

$$F_{g_pha} = Conv_{1\times1}\left(Cat\left(F_{msi_pha}, F_{hsi_pha}\right)\right) \tag{6}$$

$$F_g = IDFT\left(F_{g_amp}, F_{g_pha}\right) \tag{7}$$

where $IDFT(\bullet)$ represents the IDFT, $Cat(\bullet)$ denotes concatenation along the channel dimension, and $Conv_{1\times1}$ denotes 1×1 convolutional block. Owing to the ability of the Fourier transform to capture global image information, this process facilitates the fusion of global energy and structural information from HSI and MSI features under a large receptive field.

However, global frequency-domain fusion inherently emphasizes low-frequency components and coarse structural patterns, which may weaken the representation of high-frequency local variations and fine-grained texture details. To address this limitation, we introduce a local frequency-domain fusion strategy. In this approach, hyperspectral and multispectral features are partitioned into four spatial regions, and the DFT is applied to each region to extract the corresponding amplitude and phase components of local features. The entire process is described as follows:

$$F_{hsi_amp}^{p_i}, F_{hsi_pha}^{p_i} = DFT\left(F_{hsi}^{p_i}\right) \tag{8}$$

$$F_{msi_amp}^{p_i}, F_{msi_pha}^{p_i} = DFT\left(F_{msi}^{p_i}\right) \tag{9}$$

where p_i denotes the ith partition of the image feature, $i = 1, 2, 3, 4$. Operating analogously to the global fusion strategy, this stage independently concatenates the amplitude and phase components of the corresponding HSI and MSI feature patches along the channel dimension. These concatenated representations are then fused through a series of convolutional layers with LeakyReLU activations. Finally, the fused frequency-domain features for each patch are transformed back to the spatial domain via the IDFT. It can be described as follows:

$$F_{amp}^{p_i} = Conv_{1\times1}\left(Cat\left(F_{hsi_amp}^{p_i}, F_{msi_amp}^{p_i}\right)\right) \tag{10}$$

$$F_{pha}^{p_i} = Conv_{1\times1}\left(Cat\left(F_{hsi_pha}^{p_i}, F_{msi_pha}^{p_i}\right)\right) \tag{11}$$

$$F_l^{p_i} = IDFT\left(F_{amp}^{p_i}, F_{pha}^{p_i}\right) \tag{12}$$

where $F_{amp}^{p_i}$ and $F_{pha}^{p_i}$ denote the fused amplitude component and phase component of the ith partition, respectively. $F_l^{p_i}$ denotes fused result of the ith partition. The fused results of all partitions are then reassembled according to their original spatial order to obtain the local frequency-domain fusion result F_l. The F_l and the F_g are concatenated along the channel dimension and subsequently fused by multiple 3×3 convolutional layers to obtain the final global–local frequency-domain fusion result F_{gl}. This hierarchical fusion enables complementary integration of global structural consistency and local texture fidelity. The above process is described as follows:

$$F_{gl} = Conv_{3\times3}\left(Cat\left(F_l, F_g\right)\right) \tag{13}$$

where $Conv_{3\times3}$ denotes multiple 3×3 convolutional layers followed by LeakyReLU activation functions.

2) Spectral–Spatial Rectification Block: We adopt the SSRB [36] to optimize the utilization of spectral information inherent in both HSI and MSI, while mitigating spatial degradation present in both modalities. Before being fed into the SSRB, the HSI and MSI features are first concatenated along the channel dimension. The concatenated features then undergo preliminary fusion through multiple 3×3 convolutional layers followed by LeakyReLU activation functions, which effectively aggregate low-level S^2 information while preserving feature integrity. The preliminarily fused features are then fed into the SSRB for advanced S^2 interaction and rectification. The entire S^2 rectification process can be formulated as follows:

$$F_{ss} = SSRB\left(Conv_{3\times3}\left(Cat\left(F_{hsi}, F_{msi}\right)\right)\right) \tag{14}$$

where F_{ss} denotes the final result after S^2 interaction and rectification.

3) Differential S^2 Complementarity Module: The multiscale fusion strategy aggregates the preliminarily fused features to reconstruct the final image, therefore, the quality of the preliminary fusion critically affects the final fusion performance. For this reason, we design the DSSCM to

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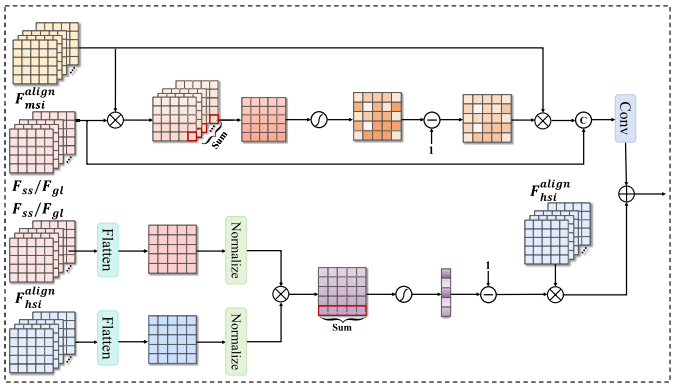


Fig. 2. Architecture of the DSSCM.

mitigate the S^2 decoupling distortions in the fused results of the two branches in SFFN. As illustrated in Fig. 2, this module enhances S^2 representations by explicitly modeling discrepancies rather than relying on direct subtraction. Specifically, pixel-level spatial differences are captured through a differential attention mechanism: we first compute spatial similarity via element-wise multiplication and channel compression, then derive attention weights from this similarity and subtract them from 1 to obtain complementary spatial differential attention values. Spectral differences occurring at the channel level are modeled similarly by compressing the spatial dimension to compute spectral similarity, followed by the same differential attention process.

In practice, to further enhance the representations of the global–local frequency-domain fusion result F_{gl} and the spectral–spatial rectification result F_{ss}, the original MSI and HSI are first processed through several 3×3 convolutional layers to achieve semantic alignment with these two-branch features. For F_{ss}, spatial information enhancement is achieved by explicitly modeling discrepancies between the MSI features and F_{ss}. We compute pixel-wise spatial similarity by performing element-wise multiplication followed by channel-wise summation. The resulting similarity map is fed into multiple convolutional layers and a sigmoid activation function to generate spatial similarity weights, which are subtracted from 1 to obtain spatial differential attention weights $w_{spa_diff}^{ss}$. The differential spatial information is then extracted by weighting the MSI features with this attention map, channel-wise concatenated with F_{ss}, and further refined by multiple 3×3 convolutional layers to produce the spatially enhanced fusion result F'_{ss}. The process is formalized as follows:

$$w_{spa_diff}^{ss} = 1 - Conv_w\left(sum_c\left(F_{ss} \times F_{msi}^{align}\right)\right) \tag{15}$$

$$F'_{ss} = Conv_{3\times3}\left(Cat\left(F_{ss}, F_{msi}^{align} \times w_{spa_diff}^{ss}\right)\right) \tag{16}$$

where $Conv_w$ denotes the weight generation block consisting of multiple 3×3 convolutions and a sigmoid activation function, F_{msi}^{align} represents the semantically aligned MSI features and sum_c denotes summation along the channel dimension.

To enhance the spectral information of F_{ss}, we first flatten the HSI features and F_{ss} into 2-D representations along the spatial dimension. After normalization, the cosine similarity between the corresponding channels of the two representations is calculated, respectively. Subsequently, the spectral similarity weights are obtained through a sigmoid activation function, and the spectral differential attention weights $w_{spe_diff}^{ss}$ are generated by subtracting the resulting similarity weights from 1. The differential spectral information is extracted by weighting the HSI features with the spectral differential attention map. Finally, the extracted differential spectral information is directly added to the F'_{ss}, generating integrated features F''_{ss} that simultaneously boost both spatial information and spectral fidelity. The process is illustrated as follows:

$$F_{r_ss} = Norm\left(Flatten\left(F_{ss}\right)\right) \tag{17}$$

$$F_{r_hsi} = Norm\left(Flatten\left(F_{hsi}^{align}\right)\right) \tag{18}$$

$$w_{spe_diff}^{ss} = 1 - CS\left(F_{r_ss}, F_{r_hsi}\right) \tag{19}$$

$$F''_{ss} = F'_{ss} + F_{hsi}^{align} \times w_{spe_diff}^{ss} \tag{20}$$

where $Flatten(\bullet)$ denotes unfolding along the spatial dimension, and F_{r_ss} and F_{r_hsi}, respectively, denote the spectral–spatial rerctification result and the semantically aligned HSI features after spatial flattening. $Norm$ denotes normalization, F_{hsi}^{align} represents the semantically aligned HSI features and $CS(\bullet)$ denotes the cosine similarity metric.

For global–local frequency-domain fusion result F_{gl}, the strategy for enhancing S^2 information follows the same principle as that adopted in F_{ss}. The corresponding process is illustrated as follows:

$$w_{spa_diff}^{gl} = 1 - Conv_w\left(sum_c\left(F_{gl} \times F_{msi}^{align}\right)\right) \tag{21}$$

$$F'_{gl} = Conv_{3\times3}\left(Cat\left(F_{gl}, F_{msi}^{align} \times w_{spa_diff}^{gl}\right)\right) \tag{22}$$

$$F_{r_gl} = Norm\left(Flatten\left(F_{gl}\right)\right) \tag{23}$$

$$w_{spe_diff}^{gl} = 1 - CS\left(F_{r_gl}, F_{r_hsi}\right) \tag{24}$$

$$F''_{gl} = F'_{gl} + F_{hsi}^{align} \times w_{spe_diff}^{gl} \tag{25}$$

where F'_{gl}, F_{r_gl}, and F''_{gl} denote the global–local frequency-domain fusion features after spatial information enhancement, spatial flattening, and S^2 information enhancement, respectively, $w_{spa_diff}^{gl}$ and $w_{spe_diff}^{gl}$ represent the spatial difference attention weights and spectral difference attention weights of F_{gl}, respectively.

4) High-Frequency Information Fusion Module: In the deep fusion stage of the network, high-frequency image information tends to be progressively attenuated. Notably, MSI typically exhibits higher spatial resolution than HSI. Therefore, we leverage the high-frequency information extracted from MSI features to enhance F''_{ss} and F''_{gl}. The Fourier transform has a strong capability to separate high- and low-frequency components of an image. Based on this property, the MSI features are first transformed into the frequency domain using the DFT. Subsequently, a high-pass filter is applied to the frequency-domain features to extract high-frequency information, and then we convert it back to the spatial domain by IDFT to obtain the high-frequency spatial information x_h. The entire process can be formulated as follows:

$$x_h = IDFT\left(filter\left(DFT\left(F_{msi}^{align}\right)\right)\right), \tag{26}$$

where $filter(\bullet)$ denotes the high-frequency filter.

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Then, x_h is concatenated along the channel dimension with the F''_{gl} and F''_{ss}, respectively. The final fused result is then obtained through two independent 3×3 convolutional blocks. The process can be expressed as follows:

$$F^o_{gl} = Conv_{3\times3}\left(Cat\left(x_h, F''_{gl}\right)\right) \tag{27}$$

$$F^o_{ss} = Conv_{3\times3}\left(Cat\left(x_h, F''_{ss}\right)\right) \tag{28}$$

where F^o_{gl} and F^o_{ss} denote the fusion result obtained by integrating F''_{gl} and F''_{ss} with x_h, respectively.

C. Loss Function

Since our SSDAN produces outputs at multiple scales, we adopt a multiscale loss strategy. Considering that the L_1 loss has faster convergence and better robustness than other common used loss functions, so we employ L_1 loss as the model's loss function at each scale. The overall loss function is defined as follows:

$$L_1 = \frac{1}{hwb}\sum_{i=1}^{h}\sum_{j=1}^{w}\sum_{k=1}^{b}\|F_{i,j,k} - X_{i,j,k}\|_1 \tag{29}$$

$$L_{total} = \sum_{s=1}^{n} L_1^s,\ n = 3 \tag{30}$$

where h, w, and b denote the height, width, and channel size of the image, respectively, i, j, and k denote the spatial location, and L_1^s denotes L_1 loss applied to the model output at the sth scale.

IV. Experiments

A. Experimental Settings

1) Datasets: We evaluate the proposed method by conducting experiments on three common public hyperspectral imaging datasets: CAVE, Harvard, and KAIST. Across all three datasets, every HSI features 31 spectral channels, corresponding to spatial resolutions of 512×512, 1392×1040, and 2704×3376 in sequence. Specifically, for CAVE, 20 HSIs are utilized for training, with the remaining 12 reserved for testing. For Harvard, 30 HSIs constitute the training set, and the other 20 are used for testing. For KAIST, the training set comprises 20 HSIs, while the remaining 10 are employed for testing.

2) Data Simulation: The training of supervised methods requires paired LR-HSIs, HR-MSIs, and corresponding HR-HSIs, which are unavailable in real scenarios. Therefore, the original HSIs are used as the ground truth (GT), and the required inputs are synthesized through simulation. LR-HSIs are generated by first convolving the GT with an 8×8 Gaussian kernel, then downsampling by a factor of 8. HR-MSIs were synthesized by applying the Nikon D700 camera's spectral response function to the GT. For patch extraction, a 64×64 size is used throughout: on the CAVE dataset, GT and HR-MSIs are cropped with a stride of 32, and LR-HSIs with a stride of 4, on the Harvard and KAIST datasets, the corresponding strides are 64 and 8.

3) Implementation Details: All methods are trained on an A800 GPU, with PyTorch version 1.10.1. In our method, four SFFNs are employed in the encoding stage. The Adam optimizer and a multiscale L_1 loss are used to minimize the reconstruction error between the recovered HR-HSI and the GT. Training is conducted for 800 epochs with a cosine annealing learning rate schedule. The training and testing datasets remain consistent across all methods.

B. Comparison Methods

To validate the superiority of our SSDAN, comparisons are conducted with several classical HSI fusion methods and recent SoTA approaches, including SSR-Net [37], LGCT [38], MIMO-SST [35], LRTN [39], SFIGNet [40], and OTIAS [41]. For all compared methods, the dataset partitioning and data preprocessing protocols are kept consistent with those described in Section IV-A. For each dataset, the hyperparameter configurations of the comparative methods are strictly follow those reported in their original publications to ensure a fair comparison.

C. Evaluation Indicators

To quantitatively assess the proposed method, we employ five standard image quality metrics. Specifically, we use: 1) peak signal-to-noise ratio (PSNR), where higher values denote greater fidelity to the reference; 2) spectral angle mapper (SAM), where smaller angles indicate lower spectral distortion; 3) erreur relative globale adimensionnelle de synthése (ERGAS), a comprehensive radiometric error measure where lower scores reflect higher quality; 4) structural similarity index measure (SSIM), which evaluates perceptual structural similarity, with values approaching 1 being ideal; and 5) root-mean-square error (RMSE), where smaller errors correspond to more accurate reconstruction.

D. Results on the CAVE Dataset

Fig. 3 presents the visual reconstruction results on the CAVE dataset, where pseudo-color images are generated from the 9th, 19th, and 29th spectral bands for display. These visualizations facilitate a comparative analysis between our SSDAN and the competing methods. Specifically, the reconstructed images are shown in the first and third rows, while the second and fourth rows depict the corresponding error maps relative to the GT. To enable a more detailed examination of representative image features, critical regions are highlighted using red boxes and magnified. As shown in the local zoomed-in view area in Fig. 3, our SSDAN preserves more distinct texture details than the competing methods. Moreover, the residual maps indicate that the deviation between our reconstructed results and the GT is minimal among all evaluated methods. Table I presents the mean values of each objective evaluation metric, computed specifically on the test set of the CAVE dataset. The experimental results illustrate that the proposed SSDAN achieves optimal performance compared to other SoTA methods across all evaluation metrics. Notably, our SSDAN achieves a

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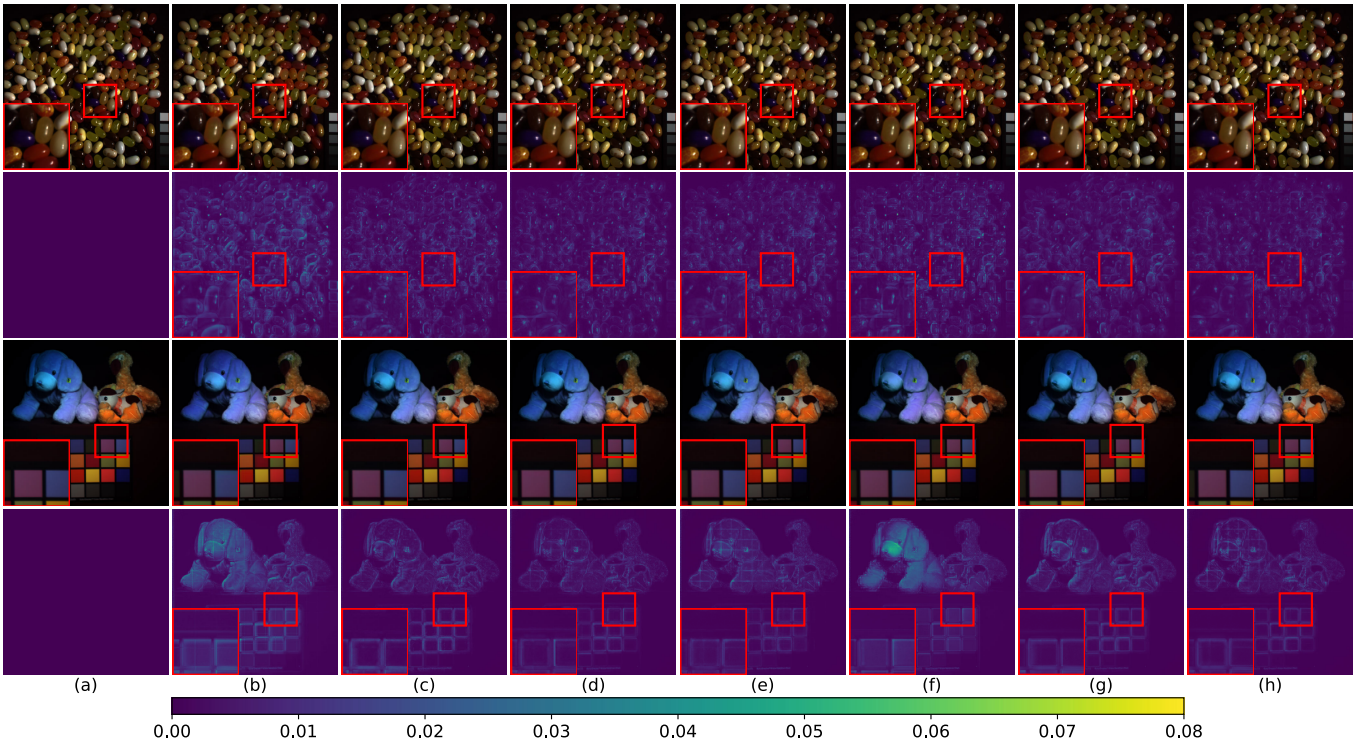


Fig. 3. True-color representations of the "jelly_beans" and "stuffed_toys" scenes are shown in rows one and three, while row two and four depict their respective residual maps comparing the fused outputs to the GT. (a) GT. (b) SSR-Net. (c) LGCT. (d) MIMO-SST. (e) LRTN. (f) SFIGNet. (g) OTIAS. (h) Ours.

TABLE I
QUANTITATIVE INDEXES OF THE TEST METHODS ON CAVE DATASET (BOLD: OPTIMAL AND UNDERLINED: SUBOPTIMAL)

Method	PSNR	SAM	RMSE	SSIM	ERGAS
SSR-Net	44.0194	4.1589	2.0438	0.9886	0.8802
LGCT	47.7472	2.5469	1.3267	0.9935	0.5830
MIMO-SST	<u>48.3056</u>	2.5226	<u>1.2439</u>	0.9939	<u>0.5493</u>
LRTN	47.9861	<u>2.3337</u>	1.2891	<u>0.9942</u>	0.5576
SFIGNet	45.7255	2.8897	1.6942	0.9924	0.7212
OTIAS	47.8902	2.4457	1.3020	<u>0.9942</u>	0.5611
Ours	**49.0519**	**2.2188**	**1.1347**	**0.9949**	**0.4997**

TABLE II
QUANTITATIVE INDEXES OF THE TEST METHODS ON HARVARD DATASET (BOLD: OPTIMAL AND UNDERLINED: SUBOPTIMAL)

Method	PSNR	SAM	RMSE	SSIM	ERGAS
SSR-Net	45.9679	3.1708	1.9215	0.9830	1.2396
LGCT	47.2366	2.8867	1.7338	0.9846	1.0297
MIMO-SST	47.1917	2.9307	1.7317	0.9846	1.0257
LRTN	47.2505	2.8974	1.7115	<u>0.9847</u>	1.0252
SFIGNet	46.7870	2.9748	1.7896	0.9843	1.0711
OTIAS	<u>47.3901</u>	**2.8248**	<u>1.7090</u>	**0.9849**	<u>1.0032</u>
Ours	**47.4242**	<u>2.8470</u>	**1.6968**	**0.9849**	**0.9931**

significant improvement in PSNR, surpassing the second-best method by 0.7463 dB. These results convincingly demonstrate that SSDAN achieves higher reconstruction fidelity and superior image quality.

E. Results on the Harvard Dataset

The experimental results of quantitative analysis on the Harvard dataset are shown in Table II. It can be observed that our SSDAN attains the optimal performance in terms of PSNR, RMSE, SSIM, and ERGAS, and the second-best performance in SAM, while remaining highly comparable to the optimal result. However, except for SSR-Net and SFIGNet, the performance discrepancies among the other methods are relatively minor, which may be ascribed to the large scale of the dataset. The large dataset provides sufficient training samples for all methods to effectively capture the intrinsic characteristics of the images. A visual scrutiny of the reconstructed images in Fig. 4 shows no obvious differences among the tested methods. However, inspection of the branch part in the locally enlarged regions indicates that the reconstructed images produced by our SSDAN are significantly clearer than the other methods. Moreover, further analysis of the fine-grained structures in the residual maps shows that the outputs of SSDAN are more consistent with the GT. Overall, our SSDAN achieves superior performance in preserving spatial texture details and recovering sharp edges.

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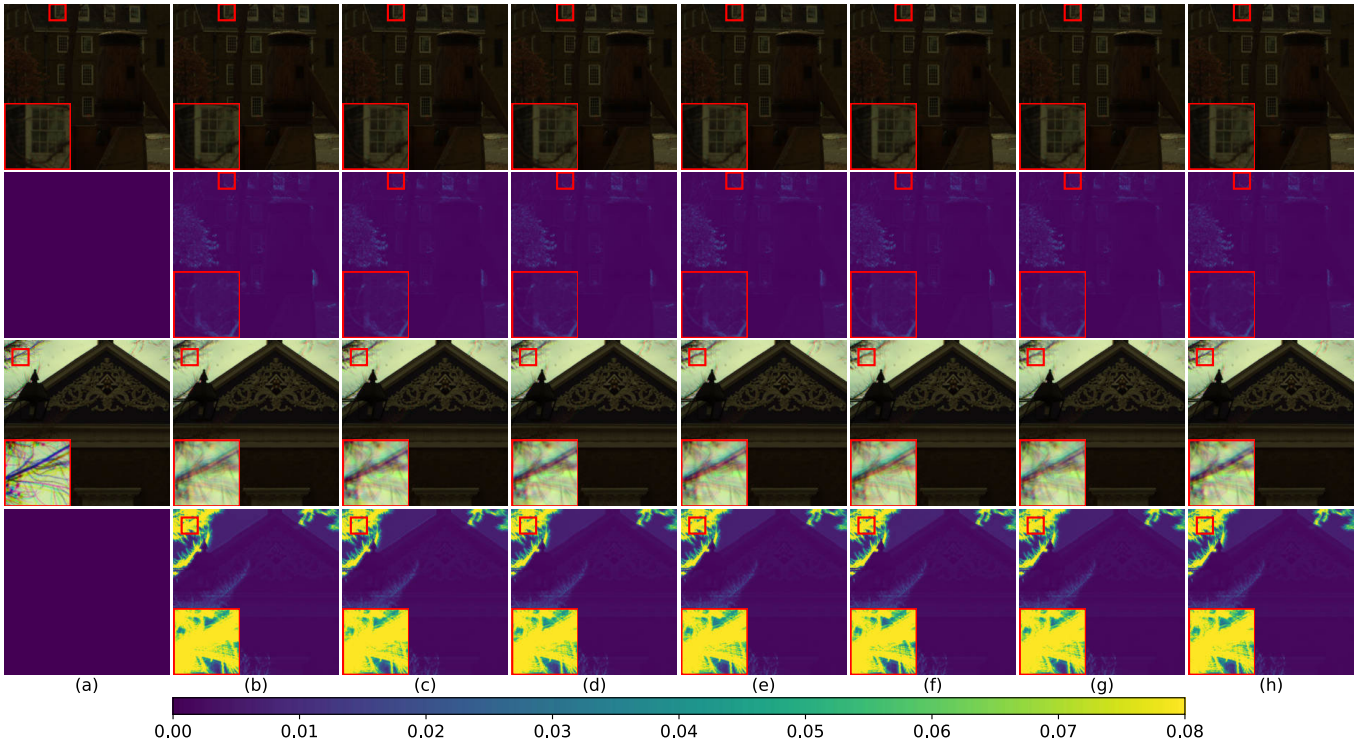


Fig. 4. True-color representations of the "imge1" and "imge2" scenes are shown in rows one and three, while row two and four depict their respective residual maps comparing the fused outputs to the GT. (a) GT. (b) SSR-Net. (c) LGCT. (d) MIMO-SST. (e) LRTN. (f) SFIGNet. (g) OTIAS. (h) Ours.

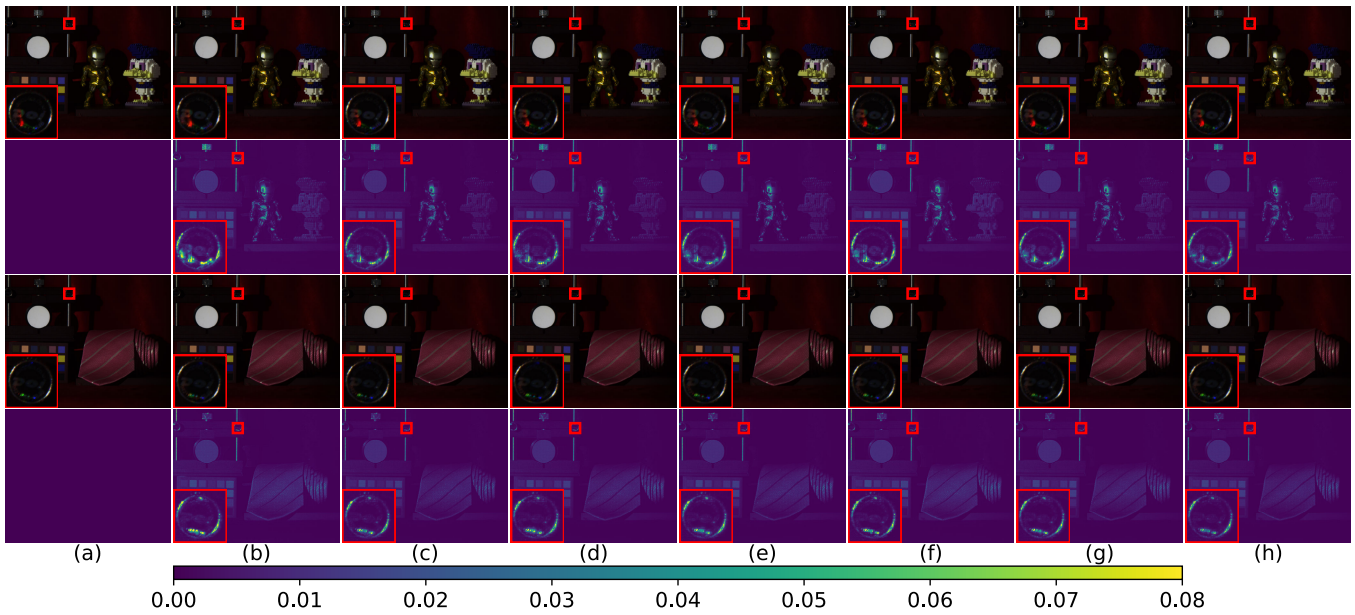


Fig. 5. True-color representations of the "scene21_reflectance" and "scene22_reflectance" scenes are shown in rows one and three, while row two and four depict their respective residual maps comparing the fused outputs to the GT. (a) GT. (b) SSR-Net. (c) LGCT. (d) MIMO-SST. (e) LRTN. (f) SFIGNet. (g) OTIAS. (h) Ours.

F. Results on the KAIST Dataset

As shown in Table III, our SSDAN outperforms all competing methods and achieves the optimal performance across all evaluation metrics. Moreover, the performance gap is more distinct than that observed on the Harvard dataset, indicating that SSDAN maintains strong performance on large-scale scene images. This superior performance stems mainly from the simultaneous fusion in both global and local frequency domains, which enables more effective preservation of the inherent structural consistency of the images. In terms of qualitative results, visual scrutiny of the reconstructed

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TABLE III
QUANTITATIVE INDEXES OF THE TEST METHODS ON KAIST DATASET (BOLD: OPTIMAL AND UNDERLINED: SUBOPTIMAL)

Method	PSNR	SAM	RMSE	SSIM	ERGAS
SSR-Net	43.7032	3.2364	1.7760	0.9854	1.3677
LGCT	<u>45.6348</u>	<u>2.4921</u>	<u>1.4091</u>	<u>0.9887</u>	<u>1.1074</u>
MIMO-SST	45.5200	2.5701	1.4315	0.9885	1.1218
LRTN	45.3389	2.6545	1.4531	0.9881	1.1429
SFIGNet	44.3001	2.8416	1.6497	0.9874	1.2971
OTIAS	45.5893	2.5069	1.4200	<u>0.9887</u>	1.1083
Ours	**45.8791**	**2.4655**	**1.3743**	**0.9890**	**1.0758**

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF ABLATION ON THE CAVE DATASET

Method	PSNR	SAM	RMSE	SSIM	ERGAS
w/o SFFN	48.4218	2.3374	1.2275	0.9942	1.2275
w/o DSSCM	48.8468	2.2608	1.1649	0.9946	0.5120
Ours	**49.0519**	**2.2188**	**1.1347**	**0.9949**	**0.4997**

TABLE V
HYPERPARAMETER EXPERIMENTS ON NETWORK DEPTH

Method	PSNR	SAM	RMSE	SSIM	ERGAS
N=2	48.7909	2.2832	1.1739	0.9946	0.5176
N=3	48.8163	2.2630	1.1718	0.9946	0.5134
N=4	**49.0519**	**2.2188**	**1.1347**	**0.9949**	**0.4997**
N=5	49.0300	2.2202	1.1355	0.9948	0.5027

images in Fig. 5 shows that SSDAN produces clearer and more detailed representations. Meanwhile, the residual maps exhibit more pronounced purple regions, demonstrating that the reconstructed images produced by our SSDAN have a higher similarity to the GT.

G. Ablation Study

This section presents an ablation study to evaluate the contribution of the proposed SFFN and DSSCM. All experiments are conducted on the CAVE dataset under consistent settings, with performance assessed using five standard image quality metrics. To evaluate the effectiveness of each module, we conduct an ablation study in which the SFFN is replaced with a standard residual convolutional block, and the DSSCM is removed. The results are shown in Table IV, where "w/o SFFN" denotes replacing the SFFN with a standard residual convolutional block and "w/o DSSCM" denotes the removal of the DSSCM. Compared with the baseline model, introducing only the SFFN (without DSSCM) improves the PSNR by approximately 0.4 dB, while further incorporating the DSSCM yields an additional improvement of approximately 0.2 dB. These results demonstrate that both SFFN and DSSCM make substantial contributions to improving the reconstruction quality of HSIs.

H. Hyperparameter Experiments

To evaluate the impact of network depth, we assess the performance of SSDAN using different numbers of stacked SFFN (denoted as N). The results are shown in Table V for N ranging from 2 to 5. It shows a significant performance improvement when increasing the N from 2 to 4, with PSNR increasing by 0.2610 dB. However, when N is further increased to 5, there is a slight drop in PSNR, accompanied by an increase in computational cost. Therefore, N = 4 is selected as the optimal choice.

V. CONCLUSION

In this article, we propose an SSDAN for HSI fusion. The proposed network achieves high-quality image reconstruction by performing multidimensional enhancement of HSI–MSI fusion features across both spatial and frequency domains. Specifically, SSDAN first integrates spatial- and frequency-domain features to enable comprehensive global information capture while preserving fine-grained local texture details. Subsequently, we design a DSSCM and an HFIFM to further refine the spatial and spectral characteristics of the fused output. This dual-module architecture aims to preserve spatial texture details to the greatest extent while alleviating S^2 decoupling distortions. Experimental results demonstrate that the proposed method outperforms existing SoTA approaches in terms of both quantitative and qualitative experimental results.

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